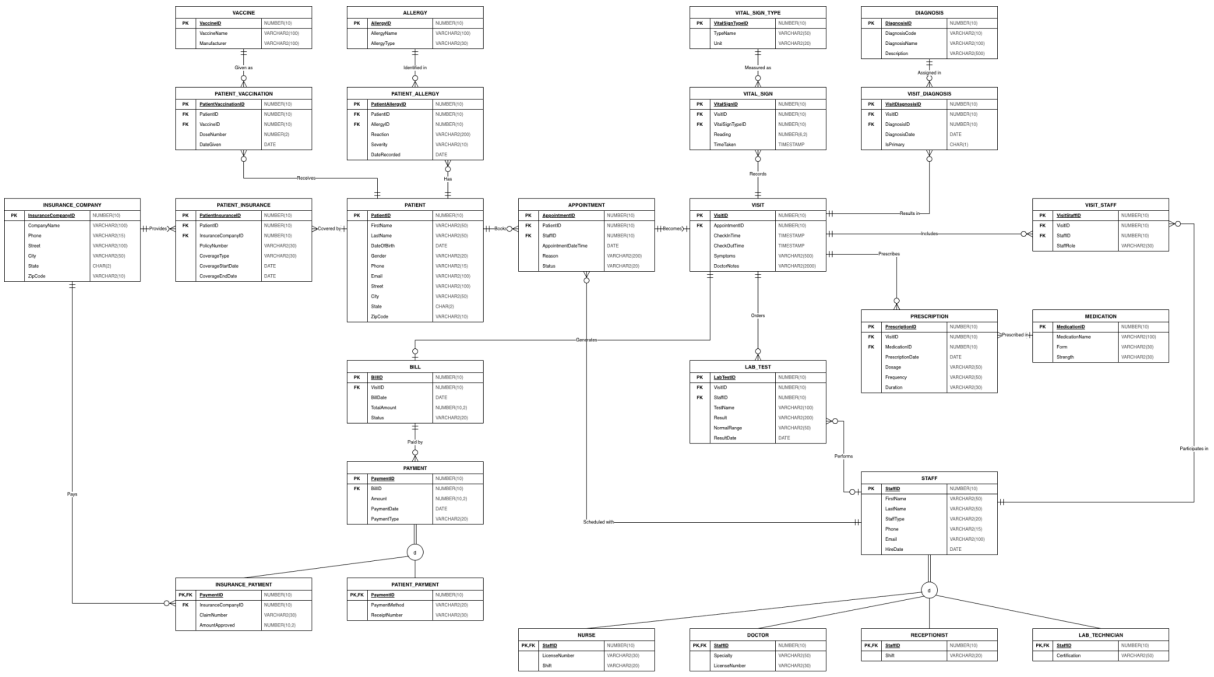
Massachusetts Health Network Database
DAMG 6210: Data Management and Database Design

GitHub Repository: https://github.com/harpereskey/DAMG6210

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Logical ERD:



Revisions:

- We separated reference data from patient records: Information such as vaccine names, allergy types, vital sign types, and diagnosis codes would have been repeated on every record. We moved it into reference entities (VACCINE, ALLERGY, VITAL_SIGN_TYPE, DIAGNOSIS), and each patient or visit record

(PATIENT_VACCINATION, PATIENT_ALLERGY, VITAL_SIGN, VISIT_DIAGNOSIS) now points to them with a foreign key:

- We resolved all many to many relationships with associative entities.
    - PRESCRIPTION links VISIT and MEDICATION and stores the prescription date, dosage, frequency, and duration.
    - PATIENT_INSURANCE links PATIENT and INSURANCE_COMPANY and stores the policy number, coverage type, and coverage start and end dates.
    - VISIT_STAFF links VISIT and STAFF and records the staff member's role during the visit.
    - PATIENT_VACCINATION, PATIENT_ALLERGY, VISIT_DIAGNOSIS, and VITAL_SIGN resolve the remaining many to many relationships between patients or visits and the reference entities above.
- We decomposed composite attributes: Addresses are split into Street, City, State, and ZipCode. The appointment's separate Date and Time attributes are stored as a single AppointmentDateTime of type TIMESTAMP, which matches CheckInTime and CheckOutTime on VISIT.
- We added keys, data types, and constraints: Every entity has its own primary key, and foreign keys implement each relationship. All attributes have Oracle data types. Subtypes (DOCTOR, NURSE, RECEPTIONIST, LAB_TECHNICIAN, INSURANCE_PAYMENT, PATIENT_PAYMENT) use the supertype's ID as both primary key and foreign key, and StaffType and PayerType act as the discriminators. Unique and check constraints that the notation can't show, such as one visit per appointment and one primary diagnosis per visit, are listed in a note on the diagram.
- We tightened relationships to enforce business rule: Appointments are scheduled with a DOCTOR and lab tests are performed by a LAB_TECHNICIAN, instead of any STAFF member. INSURANCE_PAYMENT now references PATIENT_INSURANCE, so each claim is tied to the specific policy it was paid under.

Denormalization:

- City and State are stored with ZipCode. Strictly, ZipCode determines City and State, which is a transitive dependency. We kept them together because a separate ZIP lookup table would add a join to every address query for very little benefit.
- LAB_TEST stores NormalRange with each result. The range depends on the test, but it can change over time and between labs, so storing it with the result preserves the range that applied when the test was run.
- BILL stores Status: Whether a bill is paid can be derived by comparing TotalAmount to the sum of its payments. We store it directly so the billing status can be checked without recalculating payments each time.